

The Student Research Lead Handbook (retired this edition, kept on file)

HONR 46400 · Evidence-Driven Research

⚠️ Retired for the Fall 2026 edition: kept for a future one

The Student Research Lead role does not run this semester, so nothing in this document is required of you. Every Monday and Wednesday lecture from Week 2 now opens with a ten-minute **Lab Meeting** instead, and it is an **open round**: the instructor asks the room how the projects are going, and the room answers. What you decided since last time, what your evidence looks like, where somebody is stuck. **Nobody is assigned to report**, on any lecture. There is **no slot draw**: the assignment that once named your dates is withdrawn: **nothing is prepared** before class, by anyone, and **nothing said in the lab meeting is graded**. The instructor leads the lesson from minute 10 and owns accuracy, the AI tooling and the clock.

The **25% Student Research Lead** grade category retired with the role. In its place, **Lecture Notebooks (20%)** collects the notebook you work in class, once a week. Everyone hands one in, and it is graded on completion alone.

Nothing below has been removed. This document is kept whole for a future edition that reinstates the role. Where a procedure would mislead you today, it carries a 🚫 **Retired procedure** note naming what happens instead. (*Rulings: [_project_docs/DECISIONS.md](#), D74 and D75, 2026-08-31. D75 withdrew the assigned reporter and the slot draw, and removed the injected 📌 My Report Plan cell from every notebook; the lecture's own cell is now simply ### 📌 Lab Meeting.*)

Your guide to leading a class the way a researcher leads an investigation.

What runs this semester instead: the Lab Meeting

Everything after this section describes the retired Student Research Lead role and is kept whole for a future edition. This section is what is actually true in Fall 2026.

Every Monday and Wednesday lecture from Week 2 opens with a **ten-minute lab meeting**, and it is an **open round**. Nobody is assigned to it, on any lecture. There is no draw, no list of dates, and nothing to get ready the night before.

- **Minutes 0–10: the instructor asks the room how the projects are going.** You answer when you have something to say: what you decided since last time, what your evidence is starting to look like, where you are stuck. You speak from your own project, the one thing in the room you have actually been inside, so you are never asked to explain unfamiliar material to your classmates.
- **Questions run both ways in those ten minutes.** When a classmate says where they are, [socratic_question_bank.md](#) in this folder is where to find the question worth asking back.
- **From minute 10: the instructor leads the lesson.** Accuracy, the AI tooling and the clock are the instructor's, not yours. The 🧩 Research Puzzle that used to open the lecture now folds into the front of the investigation block, opened by the instructor.

Nothing said in the lab meeting is graded. There is no live rubric, nothing to submit in advance, no slot with your name on it, and no penalty attached to how the ten minutes go. It is unpressured rehearsal for the poster presentation and the Expo in November, which are graded.

What is graded instead: Lecture Notebooks (20%)

Category	Weight
Attendance	1%

Category	Weight
Participation	9%
IYT Practice	15%
Lecture Notebooks	20%
Final Project	55%
Total	100%

Lecture Notebooks (20%) is a completion contract, the same shape as Participation and IYT Practice. One submission per week, and everyone hands one in: the notebook `nbNN` you worked in class, due **11:59 PM on the Sunday that ends that studio week** (Week 16 closes Friday Dec 11, the last class day). It is graded on **completion only**: worked through and handed in: never on whether your answers came out right, and never on anything you did or did not say in a lab meeting. Credit is **1.0** on time, **0.5** within seven days, **0** after that. Across the **16** weeks the lowest **2** credits drop automatically, so

```
lecture notebook points = 20.0 × (sum of your highest 14 credits) / 14
```

The dated list lives in `planning/LECTURE_NOTEBOOK_SCHEDULE.md`, and your course platform carries the same dates. Where the platform and this repository disagree on a date, the platform is right.

Lecture Notebooks is **not** participation. Participation's ± 0.9 -point contribution adjustment belongs to the 9% block alone and never touches this one.

What a Student Research Lead is

For most Monday and Wednesday lectures this semester, one of you runs the room. That person is the **Student Research Lead**, or **SRL** for short. Your lecture slots are assigned at the start of the semester, so you will know yours early.

Being the SRL is not giving a presentation. You are not there to summarize a reading, walk through slides, or tell the class what the notebook says. You are there to run a **Socratic investigation**: a class session built out of good questions, where the room works out the ideas by thinking, committing, testing, and arguing. "Socratic" just means you lead by asking, not by telling. The best SRL sessions feel less like a lecture and more like a lab meeting where a sharp question is on the table and everyone is trying to crack it.

Here is the difference in one line. A presenter stands at the front and transfers what they know. An SRL stands at the front and makes the room think.

The philosophy: AI is your arm, not your brain

This course runs on one idea:

AI is your arm and your research assistant. It is not your brain.

An arm is powerful. It reaches, fetches, lifts, and drafts. But it does not decide where to go. You do. When you lead, you model exactly that relationship for the room. You make the class think first, commit to an answer, then use AI on purpose, and then interrogate whatever it hands back. **Interrogate** here means to question a claim hard before you trust it: ask where it came from, whether it is true, and what it is quietly assuming.

Four habits sit under everything an SRL does:

1. **You make the room think.** The class does the reasoning. Your questions do the steering.
2. **You make the room commit.** Before anyone opens an AI tool, the class writes down what it believes. A **commitment** is a stated answer you put on record before you consult AI, so you can tell later whether the AI changed your mind or just agreed with you.

3. **You consult AI strategically.** You do not ask the AI to think for the room. You point it at a specific job, then bring the answer back for inspection.
4. **You interrogate what comes back.** No answer from an AI tool ends the discussion. A human decision ends the discussion.

The everyday shorthand for the whole loop is **Ask, Verify, Document**. Ask the AI for something specific. Verify what it gives you against real evidence. Document what you decided and why. You will run the room through that loop live.

The two lead formats 🚫 (retired procedure)

🚫 **Retired minute frames.** The two tables below give the pre-D74 openers, in which the lead owned the first block. The frames now run **Monday 10 / 21 / 12 / 7** and **Wednesday 10 / 20 / 12 / 8**: the ten-minute lab meeting takes the opening block and the instructor leads from minute 10. The later boundaries are unchanged (31–43 and 43–50 on Monday, 30–42 and 42–50 on Wednesday), so the descriptions of those blocks below still hold.

Each of your slots is either a Monday or a Wednesday, and each has its own shape. Both run 50 minutes. The minute frames below are fixed, and the instructor posts checkpoint signals at the section boundaries so you always know your pace.

Monday: the guided investigation

Monday is where a new idea gets built from a puzzle. The class starts from curiosity, commits to a belief, investigates with AI as the research partner, and lands on a decision it can defend.

Minutes	Section	What you do
0–9	Your research puzzle	Open with one concrete puzzle. Elicit prior beliefs. Get the room to commit to an answer in writing before any AI is opened.
9–31	Guided AI investigation	You facilitate as the class uses its AI research partner on the puzzle. You direct the prompts, run a human-versus-AI comparison, and probe what comes back. The instructor monitors for accuracy in the background.
31–43	Human verification and formalization	The instructor steps in to formalize the concept and lock down what is actually correct. You help connect it back to the class's committed answers.
43–50	Decision and defense	You close by making the room commit to a decision and defend it. The class records an AI Research Ledger line and a Claim Ticket.

Your two big blocks are **0–9** (you own it completely) and **9–31** (you run the investigation while the instructor watches accuracy). The **31–43** formalization is the instructor's block; your job there is to hand off cleanly and keep the thread visible.

Wednesday: the applied AI laboratory

Wednesday is where the idea gets used under pressure. You open with a challenge, the class works hands-on with AI, then defends the results against hard questions and carries the skill into their own projects.

Minutes	Section	What you do
0–7	Your retrieval challenge	Open with a challenge that forces recall and exposes a fault line. Good challenges: a common error, an ambiguous claim, a flawed

Minutes	Section	What you do
		AI output, a design conflict, or a result that needs interpreting.
7–30	Applied AI laboratory	You steer the room through hands-on work with AI. Everyone runs prompts, splits roles, and hunts for AI-failure patterns. This is the longest block, so pacing is on you.
30–38	Peer defense	The class defends its results while classmates ask adversarial questions. Adversarial here means friendly but relentless: questions designed to find the weak point, not to attack the person.
38–42	Synthesis + accuracy lock	You state the room’s conclusion and its uncertainty in one or two lines. The instructor then locks accuracy: anything that survived challenge but is wrong gets corrected before it can enter a ledger.
42–50	Transfer to projects	You close by connecting the skill to each person’s own research project. The class records an AI Research Ledger line and a Claim Ticket.

Your owned blocks are 0–7 and 7–30. The 30–38 peer defense you referee; you keep the questions coming and the answers honest. At 38–42 you synthesize, and the instructor’s accuracy lock has the last word before the ledger.

Your slots

 **Withdrawn: you hold no slots.** D75 withdrew the draw itself, so no lecture carries your name, nothing was renamed or reassigned, and there are no dates for you to look up. If Week 1 told you which meetings were yours, those slots are withdrawn; the course platform carries the correction. The section below describes the retired assignment and is kept for a future edition.

Lead slots are **assigned at random at the beginning of the semester**, so you will know which lectures are yours from the first week. Nothing rotates and nothing shifts as the term goes on: your lectures are your lectures.

Which calendar meetings you drew is posted on the **course platform** in Week 1, as the slot assignment for the whole class. That posting is the single source of truth for your dates. Check it early, then look each of your dates up on the **Schedule page of the course website**, which tells you whether the meeting is a Monday or a Wednesday, which notebook it uses, and what it covers. This handbook never prints dates on purpose, so it never goes stale.

Your SRL Lead Brief (retired procedure)

 **Renamed and re-owned.** The cell described below still opens every lecture, but it is now ### 📌 **Lab Meeting** and its run of show belongs to the instructor. The planning cell D74 injected below it, ### 📌 **My Report Plan**, was removed by D75: no notebook carries one, because nothing about the lab meeting is prepared in advance. The lecture’s question set is now ### 🗨 **Questions to Keep You Thinking**, every question kept word for word.

Every led lecture opens with an **SRL Lead Brief** at the very start of that lecture’s notebook. It is a normal, student-visible section, so the whole class can read it. Nothing is emailed to you and nothing is held back.

Your brief names the concept in play, the compass position or design in focus, a seed puzzle you can use or adapt, and the one thing the room most needs to leave understanding.

Treat the brief as a **floor, not a ceiling**. It guarantees the session has everything it needs even on a bad week, but the good sessions are the ones where the lead adds something. Swap in an example from your own project or your own field. Restage the puzzle with different numbers, a different case, or a demonstration the room can watch. Cut a question you find flat and write a sharper one. Bring your own voice. The only fixed parts are the minute frame and the moves the rubric grades; everything else is yours to make interesting.

Your preparation timeline 🛑 (*retired procedure*)

🛑 **The notebook is no longer submitted the day before.** You now hand in the week's notebook on completion, at **11:59 PM on the Sunday that ends the studio week**, under **Lecture Notebooks (20%)**: not the day before one person's lecture, and on the same terms for everyone. For the lab meeting itself there is nothing to prepare at all: no submission, no plan, no assigned question. You bring whatever your project has actually been doing. The rhythm below is the retired SRL preparation timeline, kept for a future edition.

Good SRL sessions are built, not improvised. Here is the rhythm.

About a week before your lead

- Read your **SRL Lead Brief** at the top of that lecture's notebook, then read the rest of the notebook and the matching chapter yourself, as a learner. You cannot lead an investigation into an idea you have not sat with.
- Pick your puzzle. Use the seed, sharpen it, or bring your own. A puzzle works when it has a real answer, more than one tempting wrong answer, and can be stated in a few sentences.
- Decide what **you** are adding that the brief does not have. One thing is enough: your own example, your own staging, your own opening question.
- Rough out your three Socratic questions and the moment you will have the room commit before touching AI.

The day before your lead

- Submit **your lecture's notebook, filled in**. Open the notebook for the lecture you lead, complete the 🖋 **My Lead Plan** cell at the top of your lecture, and work the rest of the notebook as a learner: the puzzle answer, the predictions, the evidence readings, the practice, the closing moves. Hand in the whole notebook on the course platform.
- 🖋 **My Lead Plan** is the part about leading rather than learning: how you will stage the puzzle, the one thing you are adding, your commitment question, the decision you will close on, and your fallback if the AI tool is down. Five lines are enough.
- The instructor reviews the notebook and sends notes on it. Treat those notes seriously; they are the difference between a session that lands and one that stalls.
- [srl_prep_template.md](#) is optional. Reach for it only if you want a longer planning worksheet than the five lines the notebook asks for.

The day of your lead

- Arrive a few minutes early. Open Colab, confirm your AI tool responds, and paste your first prompt into a scratch cell so it is ready.
- Load your puzzle where the room can see it.
- Have your fallback ready in case the AI tool is slow or down. A dead tool should never kill your session.
- Breathe. You are not performing. You are hosting a good argument.

The ten SRL moves

These are the moves that turn a session from a talk into an investigation. You will not use all ten every time. Learn them, then reach for the ones your puzzle needs.

1. **Introduce a concrete puzzle.** Start with something specific and real, not a definition. “Here are two studies that disagree about one arrow in a diagram” beats “Today we will discuss confounding.”
2. **Ask before you define.** When a term comes up, ask the room what they think it means before you or the AI supplies the definition. Their guesses tell you where the confusion lives.
3. **Elicit prior beliefs.** Get answers on the record before any evidence arrives. “Hands up, who thinks the effect is positive?” A **prior belief** is what someone expects to be true before they look, and surfacing it makes learning visible.
4. **Require commitment before AI.** Have everyone write down their answer before the AI is opened. This is the single most important move. Without it, the AI’s answer becomes everyone’s answer and no one learns anything.
5. **Direct strategic AI use.** Point the AI at one specific job with a prompt you chose in advance. Never let the room type “explain this topic” and read the result aloud. That is outsourcing the thinking.
6. **Compare human versus AI answers.** Put the room’s committed answer next to the AI’s answer, side by side. Where they agree, ask why. Where they differ, ask who is right and how you would know.
7. **Probe assumptions.** For any claim, human or AI, ask what has to be true for it to hold. An **assumption** is a condition a claim quietly depends on. Naming it is often the whole lesson.
8. **Demand counterexamples and evidence.** Ask for a case where the claim would fail, and ask what evidence would settle it. A claim with no possible counterexample and no supporting evidence is not yet a finding.
9. **Call out unjustified AI confidence.** When the AI sounds certain, ask the room to check whether the certainty is earned. Confident tone is not evidence. This is a graded habit, not a nicety.
10. **Close by making the room defend a decision.** End on a human decision, and make someone defend it out loud against a hard question. The session is not over when the AI answers. It is over when a person commits and holds up.

When an answer is wrong or uncertain

Someone will give a wrong answer. The AI will give a wrong answer. You will not always know the right one in the moment. All three are normal, and all three are useful.

- **Treat every wrong answer as data, never as failure.** A wrong answer tells the room exactly which assumption to examine. Say “interesting, walk me through how you got there,” not “no, that’s wrong.” You are studying the reasoning, not scoring it.
- **Never humiliate.** A room that fears looking foolish stops talking, and a silent room cannot be led. Protect the person, examine the idea.
- **When you are genuinely unsure, say so and turn it into an investigation.** “I don’t know either. How could we find out?” is a strong SRL move, not a weak one. Uncertainty is the raw material of research.
- **When the AI is uncertain or hedges, notice it out loud.** Missing certainty is one of the AI-failure patterns you are hunting. Do not let a vague answer pass as a settled one.

How the instructor will step in

You are not alone at the front. The instructor is monitoring, especially during the Monday investigation block, and will step in on purpose. Expect it, and do not read it as a rescue.

- If a **conceptual error starts spreading**, the instructor flags it without taking the room from you. You keep leading; the error gets corrected.
- If the **room goes quiet**, the instructor may seed a cold-call to restart the flow. Pick the thread back up.
- If the **AI produces a failure you miss**, the instructor will often ask you to ask the room about it, rather than answering directly. Follow that thread.
- The instructor owns the **Monday 31–43 formalization block**, where the correct version of the concept gets locked in.

The full details are in the instructor intervention protocol, which you do not need to read. You just need to know that stepping in is part of the design, and it keeps your session accurate.

Logistics this semester, in one place

- **You hold no slots.** Nobody is assigned to the lab meeting, on any lecture, so there is no draw and no list of dates to check.
- **You prepare nothing for it**, in advance or otherwise. When the instructor asks how the projects are going, you say where yours actually stands.

- You hand in **the week's lecture notebook**, worked through, by **11:59 PM on the Sunday that ends that studio week** (Week 16 closes Friday Dec 11). That is **Lecture Notebooks (20%)**, and it is graded on completion only.
 - Each session still ends with the class recording an **AI Research Ledger** line and a **Claim Ticket**, the two records that travel with every session in this course.
 - **Nothing said in the lab meeting is graded**, so the SRL rubric in this folder is not applied and the peer feedback form is not collected. Whatever you hear back happens live, inside the ten minutes themselves.
-

Logistics, in one place (retired procedure)

- Your slots are **assigned at random at the start of the semester**, and the assignment is posted on the **course platform** in Week 1. Look your dates up on the **Schedule page** for the format and the notebook.
 - Your **SRL Lead Brief** is already in the lecture notebook, at the very start. Start reading it and the notebook **about a week ahead**.
 - You submit **your lecture's notebook, filled in, the day before**, with its  **My Lead Plan** cell complete; the instructor sends notes on it.
 - Each session ends with the class recording an **AI Research Ledger** line and a **Claim Ticket**, the two records that travel with every session in this course.
 - Your leading is graded on the **SRL rubric** ([srl_rubric.md](#)). Read it before your first slot so you know what strong leading looks like.
 - Classmates give you quick feedback after each session on the **peer feedback form** ([srl_peer_feedback_form.md](#)). Read it. It is the fastest way to get better before your next slot.
-

You are going to be nervous the first time. Every researcher is nervous the first time they run a room. The fix is not to know every answer. The fix is to ask better questions than anyone expected, and to make the room glad it had to think. Go make them think.